

Honours Multivariate Analysis

Lecture 6 - MANOVA

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Course Outline

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 - ① Visualization and Summary Statistics [†]
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 - ③ The Multivariate Normal Distribution [†]
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 - ② Canonical Correlation Analysis §

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MANOVA overview

So far we have been investigating the mean vector of **A** population based on **ONE** sample of size n from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$: $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$.

But what if we want to compare this mean vector to that of another population, or sub-group from the same population? Enter Multivariate Analysis of Variance (MANOVA).

Suppose we measure p variables from g different populations (or groups), by drawing a possibly different size sample from each. These random samples can be arranged as

$$\begin{array}{ll} \text{Population 1:} & \mathbf{X}_{11}, \mathbf{X}_{12}, \dots, \mathbf{X}_{1n_1} \\ \text{Population 2:} & \mathbf{X}_{21}, \mathbf{X}_{22}, \dots, \mathbf{X}_{2n_2} \\ & \vdots \\ \text{Population } g: & \mathbf{X}_{g1}, \mathbf{X}_{g2}, \dots, \mathbf{X}_{gn_g} \end{array}$$

MANOVA overview

MANOVA allows us to investigate

- 1 whether the population mean vectors are the same, and
- 2 if not, which mean components differ.

There are some assumptions:

- 1 If $\mathbf{X}_{l1}, \mathbf{X}_{l2}, \dots, \mathbf{X}_{ln_l}$ are a random sample of size n_l from a population with mean $\boldsymbol{\mu}_l, l = 1, \dots, g$, then random samples from different populations are *independent*.
- 2 All populations have a common covariance matrix, $\boldsymbol{\Sigma}$.
- 3 Each population is multivariate normal.

Before we define the MANOVA model, we will first give an overview of the univariate case when $p = 1$, i.e. ANOVA.

ANOVA

The principles described above are exactly the same in ANOVA, the only difference being that each observation in each group is a single value (one variable), instead of a vector of values (multivariate).

Let X_{lj} denote the j^{th} observation from the l^{th} group. We now want to test

$$H_0: \mu_1 = \mu_2 = \cdots = \mu_g$$

where $\mu_l = \mu + (\mu_l - \mu) = \mu + \tau_l$. We are interested in the deviations τ_l associated with the l^{th} group (treatment).

Here, μ denotes the overall mean across all groups.

ANOVA

The null hypothesis can be stated in terms of τ_l as

$$H_0: \tau_1 = \tau_2 = \cdots = \tau_g = 0.$$

The response $X_{lj} \sim N(\mu + \tau_l, \sigma^2)$ can be expressed as

$$X_{lj} = \underbrace{\mu}_{\text{overall mean}} + \underbrace{\tau_l}_{\text{treatment effect}} + \underbrace{e_{lj}}_{\text{random error}}$$

where $e_{lj} \sim N(0, \sigma^2)$.

To ensure uniqueness of parameters, use the constraint $\sum_{l=1}^g n_l \tau_l = 0$.

ANOVA

By “analyse the variance”, we mean that the deviations from observations to means are broken down within and across groups. Then we see whether some groups’ means differ more than expected according to some significance level.

Therefore, we summarise the sums of squares in an ANOVA table:

Source of Variation	Sum of Squares (SS)	Degrees of Freedom
Treatment	$SS_{tr} = \sum_{l=1}^g n_l (\bar{x}_l - \bar{x})^2$	$g - 1$
Residual (Error)	$SS_{res} = \sum_{l=1}^g \sum_{j=1}^{n_l} (x_{lj} - \bar{x}_l)^2$	$\sum_{l=1}^g n_l - g$
Total	$SS_{tot} = \sum_{l=1}^g \sum_{j=1}^{n_l} (x_{lj} - \bar{x})^2$	$\sum_{l=1}^g n_l - 1$

To test

$$H_0 : \tau_1 = \tau_2 = \dots = \tau_g = 0$$

at a given significance level α , we use the test statistic

$$F = \frac{SS_{tr}/(g-1)}{SS_{res}/(\sum_{l=1}^g n_l - g)}$$

and reject if

$$F > F_{g-1, \sum n_l - g}(\alpha)$$

Example

Example 6.7 on page 298 of Johnson & Wichern. Also see `Lecture6.R`, where all the components are manually calculated from the data and the results compared to that from the `aov` function.

ANOVA Example

```
#Form data vectors
x1 <- matrix(c(9, 6, 9), 3)
x2 <- matrix(c(0, 2), 2)
x3 <- matrix(c(3, 1, 2), 3)

#Define n's
n1 <- c(length(x1), length(x2), length(x3))

#Using aov function
X <- c(x1, x2, x3)
group <- c(rep(1, n1[1]), rep(2, n1[2]), rep(3, n1[3]))
group <- as.factor(group)
fit <- aov(X ~ group)
summary(fit)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group	2	78	39	19.5	0.00435
Residuals	5	10	2		

Another way of getting the same output:

```
anova(lm(X ~ group))
```

MANOVA model

Now, instead of measuring a single variable, we consider **vectors of observations**. The model can now be written as

$$\mathbf{X}_{lj} = \boldsymbol{\mu} + \boldsymbol{\tau}_l + \mathbf{e}_{lj}, j = 1, 2, \dots, n_l, l = 1, 2, \dots, g$$

where $\mathbf{e}_{lj} \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and with $\sum_{l=1}^g n_l \boldsymbol{\tau}_l = \mathbf{0}$.

This implies that we have the following assumptions for this model:

- Each component of the observation vector $\mathbf{X}_{lj}$ satisfies the univariate model.
- The errors for the components of $\mathbf{X}_{lj}$ are correlated.
- The covariance matrix $\boldsymbol{\Sigma}$ is the same for all populations.

MANOVA model

When decomposing the total variance to construct the MANOVA table, we consider not only the sums of squares, but also the cross-products, yielding a matrix for each component of variation:

Source of Variation	Matrix of SSP's	Deg of Freedom
Treatment	$\mathbf{B} = \sum_{l=1}^g n_l (\bar{\mathbf{x}}_l - \bar{\mathbf{x}})(\bar{\mathbf{x}}_l - \bar{\mathbf{x}})'$	$g - 1$
Residual	$\mathbf{W} = \sum_{l=1}^g \sum_{j=1}^{n_l} (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l)(\mathbf{x}_{lj} - \bar{\mathbf{x}}_l)'$	$\sum_{l=1}^g n_l - g$
Total	$\mathbf{B} + \mathbf{W} = \sum_{l=1}^g \sum_{j=1}^{n_l} (\mathbf{x}_{lj} - \bar{\mathbf{x}})(\mathbf{x}_{lj} - \bar{\mathbf{x}})'$	$\sum_{l=1}^g n_l - 1$

MANOVA testing

To test $H_0: \boldsymbol{\tau}_1 = \boldsymbol{\tau}_2 = \dots = \boldsymbol{\tau}_g = \mathbf{0}$, use Wilks' lambda,

$$\Lambda^* = \frac{|\mathbf{W}|}{|\mathbf{W} + \mathbf{B}|} = \frac{|\sum_{l=1}^g \sum_{j=1}^{n_l} (\mathbf{x}_{lj} - \bar{\mathbf{x}}_l)(\mathbf{x}_{lj} - \bar{\mathbf{x}}_l)'|}{|\sum_{l=1}^g \sum_{j=1}^{n_l} (\mathbf{x}_{lj} - \bar{\mathbf{x}})(\mathbf{x}_{lj} - \bar{\mathbf{x}})'|}$$

and reject if Λ^* is too small.

As we saw in section 5.3.1, this quantity is related to the likelihood ratio criterion.

MANOVA testing

The distribution of Λ^* is tabulated, but for a few special cases we have the following exact distributions:

No. of variables	No. of groups	Sampling distribution for MVN data
$p = 1$	$g \geq 2$	$\left(\frac{\sum n_l - g}{g - 1} \right) \left(\frac{1 - \Lambda^*}{\Lambda^*} \right) \sim F_{g-1, \sum n_l - g}$
$p = 2$	$g \geq 2$	$\left(\frac{\sum n_l - g - 1}{g - 1} \right) \left(\frac{1 - \sqrt{\Lambda^*}}{\sqrt{\Lambda^*}} \right) \sim F_{2(g-1), 2(\sum n_l - g - 1)}$
$p \geq 1$	$g = 2$	$\left(\frac{\sum n_l - p - 1}{p} \right) \left(\frac{1 - \Lambda^*}{\Lambda^*} \right) \sim F_{p, \sum n_l - p - 1}$
$p \geq 1$	$g = 3$	$\left(\frac{\sum n_l - p - 2}{p} \right) \left(\frac{1 - \sqrt{\Lambda^*}}{\sqrt{\Lambda^*}} \right) \sim F_{2p, 2(\sum n_l - p - 2)}$

MANOVA testing

When $n = \sum n_l$ is large,

$$- \left(n - 1 - \frac{(p+g)}{2} \right) \ln(\Lambda^*) \sim \chi_{p(g-1)}^2$$

and we will reject H_0 if

$$- \left(n - 1 - \frac{(p+g)}{2} \right) \ln \left(\frac{|\mathbf{W}|}{|\mathbf{W} + \mathbf{B}|} \right) > \chi_{p(g-1)}^2(\alpha)$$

Example 6.9 on page 304 of Johnson & Wichern continues the previous example.
Again see [Lecture6.R](#).

MANOVA Example

```
#Form data matrices
X1 <- matrix(c(9,6,9,3,2,7), 3)
X2 <- matrix(c(0,2,4,0), 2)
X3 <- matrix(c(3,1,2,8,9,7), 3)

#Define n's and p
n1 <- c(nrow(X1), nrow(X2), nrow(X3))
p <- ncol(X1)

#Using manova function
X <- rbind(X1, X2, X3)
group <- c(rep(1, n1[1]), rep(2, n1[2]), rep(3, n1[3]))
group <- as.factor(group)
fit <- manova(X ~ group)
summary(fit, 'Wilks')
```

	Df	Wilks	approx F	num Df	den Df	Pr(>F)
group	2	0.038455	8.1989	4	8	0.006234
Residuals	5					

We can then also investigate the individual ANOVA's.

Other MANOVA test statistics

We note that Wilks' lambda can also be expressed as a function of the eigenvalues $\lambda_1 > \lambda_2 > \dots > \lambda_s$ of $\mathbf{W}^{-1}\mathbf{B}$, where $s = \min(p, g - 1)$, such that

$$\Lambda^* = \prod_{i=1}^s \left(\frac{1}{1 + \lambda_i} \right)$$

Although Wilks' lambda is primarily used for MANOVA significance tests, the following test statistics – all of which are based on the above ordered eigenvalues and can be linked back to Hotelling's T^2 – have also been developed:

Other MANOVA test statistics

Roy's largest root:

$$\theta = \lambda_1$$

Pillai's V criterion / Pillai's trace:

$$V^{(s)} = \text{tr} [(\mathbf{W} + \mathbf{B})^{-1} \mathbf{B}] = \sum_{i=1}^s \frac{\lambda_i}{1 + \lambda_i}$$

The distributions of these two statistics are tabulated.

Lawley-Hotelling trace

$$U^{(s)} = \text{tr} (\mathbf{W}^{-1} \mathbf{B}) = \sum_{i=1}^s \lambda_i$$

which is approximately $\chi_{p(g-1)}^2$ distributed for large n .

Testing for equality of covariance matrices

We assumed that all $e_{lj} \sim N_p(\mathbf{0}, \Sigma)$ and independent with the same covariance matrix Σ for all g groups. To validate this assumption, we use Box's M test to test the hypothesis

$$H_0: \Sigma_1 = \Sigma_2 = \cdots = \Sigma_g$$

The test statistic is derived with the likelihood ratio testing method, with likelihood ratio test statistic

$$\Lambda = \prod_{i=1}^g \left(\frac{|\mathbf{S}_i|}{|\mathbf{S}_p|} \right)^{\frac{n_i-1}{2}}$$

where

$$\mathbf{S}_p = \frac{\sum_{i=1}^g (n_i - 1) \mathbf{S}_i}{\sum_{i=1}^g (n_i - 1)}$$

By calculating $-2 \ln \Lambda$, we derive the test statistic for Box's M-Test:

$$M = \left(\sum_{i=1}^g (n_i - 1) \right) \ln |\mathbf{S}_p| - \left(\sum_{i=1}^g (n_i - 1) \ln |\mathbf{S}_i| \right)$$

Testing for equality of covariance matrices

The test statistic, after applying appropriate scaling, has the following approximate sampling distribution:

$$\left[1 - \left(\sum_{i=1}^g \frac{1}{n_i - 1} - \frac{1}{\sum_{i=1}^g (n_i - 1)} \right) \frac{2p^2 + 3p - 1}{6(p+1)(g-1)} \right] M \sim \chi^2_{\frac{p(p+1)(g-1)}{2}}$$

We do note the following:

- This test is sensitive to some forms of non-normality.
- For example, it is sensitive to kurtosis for which the MANOVA tests are rather robust.
- The M-test may signal covariance heterogeneity in some cases where it is not damaging to the MANOVA tests.
- We may not wish to automatically rule out standard MANOVA tests if the M-test leads to rejection of H_0 .

Simultaneous confidence intervals

Suppose our MANOVA rejects the hypothesis of equal mean vectors... Now what? We create simultaneous confidence intervals for all combinations of $\boldsymbol{\tau}_k - \boldsymbol{\tau}_l$ using the Bonferroni approach to determine which groups are different.

Since $\boldsymbol{\tau}_k$ is estimated by $\bar{\mathbf{x}}_k - \bar{\mathbf{x}}$,

$$\hat{\tau}_{ki} - \hat{\tau}_{li} = \bar{x}_{ki} - \bar{x}_{li},$$

which denotes the difference between two independent sample means, for $i = 1, \dots, p$.

The variance of this contrast is

$\text{Var}(\hat{\tau}_{ki} - \hat{\tau}_{li}) = \text{Var}(\bar{X}_{ki} - \bar{X}_{li}) = \left(\frac{1}{n_k} + \frac{1}{n_l} \right) \sigma_{ii}$, which can be estimated by

$$\left(\frac{1}{n_k} + \frac{1}{n_l} \right) \frac{w_{ii}}{n - g}$$

where w_{ii} is the i^{th} diagonal element of $\mathbf{W}$, and $n = \sum_{j=1}^g n_j$.

Simultaneous confidence intervals

We now have p variables and $\frac{g(g-1)}{2}$ pairwise differences, yielding $m = \frac{pg(g-1)}{2}$ simultaneous confidence statements.

This results in a confidence interval for $\tau_{ki} - \tau_{li}$ of

$$\bar{x}_{ki} - \bar{x}_{li} \pm t_{n-g} \left(\frac{\alpha}{pg(g-1)} \right) \sqrt{\frac{w_{ii}}{n-g} \left(\frac{1}{n_k} + \frac{1}{n_l} \right)}$$

for all components $i = 1, \dots, p$ and all differences $l < k = 1, \dots, g$.

Example

For a thorough example of an application, see [Lecture6.R](#), based on example 6.11 on page 309 of Johnson & Wichern.

Homework exercise 6.1

Johnson & Wichern exercise 6.20 (adapted)

The tail lengths in mm (X_1) and wing length in mm (X_2) for 45 male hook-billed kites are given in `T6-11.dat`. Similar measurements for female hook-billed kites are given in `T5-12.dat`.

- a) Plot the male hook-billed data as a scatter diagram. Do any observations seem suspicious?
- b) Test for the equality of mean vectors for male and female hook-billed kites.
- c) Calculate simultaneous Bonferroni confidence intervals for $\mu_1 - \mu_2$.

Two-way ANOVA

The preceding methodology can be extended to data that can be grouped according to two factors, referred to as a two-way design. We will again first consider the univariate case.

Defining the scenario:

- Factor 1 has g levels
- Factor 2 has b levels
- We have n observations from each of the gb combinations of levels
- All of these observations are independent
- Let X_{lkr} denote the r^{th} observation at level l of factor 1 and level k of factor 2

Two-way ANOVA

The univariate two-way model is given by

$$X_{lkr} = \mu + \tau_l + \beta_k + \gamma_{lk} + e_{lkr}, \quad l = 1, \dots, g, \quad k = 1, \dots, b, \quad r = 1, \dots, n$$

with

$$\sum_{l=1}^g \tau_l = \sum_{k=1}^b \beta_k = \sum_{l=1}^g \gamma_{lk} = \sum_{k=1}^b \gamma_{lk} = 0$$

where

- μ represents an overall mean
- τ_l represents the fixed effect of factor 1
- β_k represents the fixed effect of factor 2
- γ_{lk} represents the interaction between factors 1 and 2
- $e_{lkr} \sim N(0, \sigma^2)$ are independent random variables

Two-way ANOVA table

Appropriate decomposition of sums of squares yields the following two-way ANOVA table:

Source of Variation	Sum of Squares	Deg of Freedom
Factor 1	$\sum_{l=1}^g bn(\bar{x}_{l.} - \bar{x})^2$	$g - 1$
Factor 2	$\sum_{k=1}^b gn(\bar{x}_{.k} - \bar{x})^2$	$b - 1$
Interaction	$\sum_{l=1}^g \sum_{k=1}^b n(\bar{x}_{lk} - \bar{x}_{l.} - \bar{x}_{.k} + \bar{x})^2$	$(g - 1)(b - 1)$
Residual (Error)	$\sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (x_{lkr} - \bar{x}_{lk})^2$	$gb(n - 1)$
Total	$\sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (x_{lkr} - \bar{x})^2$	$gbn - 1$

Two-way MANOVA

This design is once again extended to observations of multiple variables, where each response is a vector with p components. Again consider two factors with g and b levels respectively.

The two-way MANOVA model is defined as

$$\mathbf{X}_{lkr} = \boldsymbol{\mu} + \boldsymbol{\tau}_l + \boldsymbol{\beta}_k + \boldsymbol{\gamma}_{lk} + \mathbf{e}_{lkr}$$

with sums of squares and cross-products summarised in the following table:

Two-way MANOVA table

Source of Variation	Sum of Squares and cross-Products	Degrees of Freedom
Factor 1	$\sum_{l=1}^g bn(\bar{\mathbf{x}}_{l.} - \bar{\mathbf{x}})(\bar{\mathbf{x}}_{l.} - \bar{\mathbf{x}})'$	$g - 1$
Factor 2	$\sum_{k=1}^b gn(\bar{\mathbf{x}}_{.k} - \bar{\mathbf{x}})(\bar{\mathbf{x}}_{.k} - \bar{\mathbf{x}})'$	$b - 1$
Interaction	$\sum_{l=1}^g \sum_{k=1}^b n(\bar{\mathbf{x}}_{lk} - \bar{\mathbf{x}}_{l.} - \bar{\mathbf{x}}_{.k} + \bar{\mathbf{x}})(\bar{\mathbf{x}}_{lk} - \bar{\mathbf{x}}_{l.} - \bar{\mathbf{x}}_{.k} + \bar{\mathbf{x}})'$	$(g - 1)(b - 1)$
Residual (Error)	$\sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (\mathbf{x}_{lkr} - \bar{\mathbf{x}}_{lk})(\mathbf{x}_{lkr} - \bar{\mathbf{x}}_{lk})'$	$gb(n - 1)$
Total (corrected)	$\sum_{l=1}^g \sum_{k=1}^b \sum_{r=1}^n (\mathbf{x}_{lkr} - \bar{\mathbf{x}})(\mathbf{x}_{lkr} - \bar{\mathbf{x}})'$	$gbn - 1$

We can use the following statistics, with the specified distributions, to test for significance of the individual factors and their interaction:

Two-way MANOVA tests

Factor 1

Hypothesis:

$$H_0: \tau_1 = \tau_2 = \dots = \tau_g = \mathbf{0} \quad (\text{No factor 1 effect})$$

vs

$$H_1: \text{At least one } \tau_l \neq \mathbf{0}$$

Test statistic:

$$\Lambda^* = \frac{|SSP_{res}|}{|SSP_{fac1} + SSP_{res}|}$$

Distribution:

$$- \left[gb(n-1) - \frac{p+1-(g-1)}{2} \right] \ln \Lambda^* \sim \chi^2_{(g-1)p}$$

Two-way MANOVA tests

Factor 2

Hypothesis:

$$H_0: \beta_1 = \beta_2 = \dots = \beta_b = \mathbf{0} \quad (\text{No factor 2 effect})$$

vs

$$H_1: \text{At least one } \beta_k \neq \mathbf{0}$$

Test statistic:

$$\Lambda^* = \frac{|SSP_{res}|}{|SSP_{fac2} + SSP_{res}|}$$

Distribution:

$$- \left[gb(n-1) - \frac{p+1-(b-1)}{2} \right] \ln \Lambda^* \sim \chi^2_{(b-1)p}$$

Two-way MANOVA tests

Interaction

Hypothesis:

$$H_0: \gamma_{11} = \gamma_{12} = \dots = \gamma_{gb} = \mathbf{0} \quad (\text{No interaction effects})$$

vs

$$H_1: \text{At least one } \gamma_{lk} \neq \mathbf{0}$$

Test statistic:

$$\Lambda^* = \frac{|SSP_{res}|}{|SSP_{int} + SSP_{res}|}$$

Distribution:

$$- \left[gb(n-1) - \frac{p+1-(g-1)(b-1)}{2} \right] \ln \Lambda^* \sim \chi^2_{(g-1)(b-1)p}$$

Two-way MANOVA tests

First conduct interaction test

- If interaction effects are significant:
 - Factor effects don't have clear interpretation
 - Rather proceed with p univariate two-way ANOVAs
 - This determines whether interaction is present between some variables but not others
- If the interactions effects are negligible:
 - Test for significant factors effects
 - Focus on the contrasts in the factors 1 and factor 2 effects
 - Use Bonferroni method to determine which levels differ significantly per variable

Two-way MANOVA example

Example 6.13 on page 318 of Johnson & Wichern.

We have three variables measuring the quality of plastic film: X_1 = tear resistance, X_2 = gloss, and X_3 = opacity. $n = 5$ observations were made for each combination of 2 factors, each with two levels: Factor 1 = Change in rate of extrusion, and Factor 2 = Amount of additive. The data is given in `T6-4.dat` (also see `Lecture6.R`).

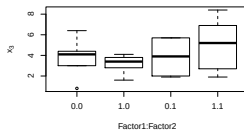
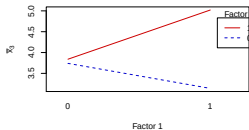
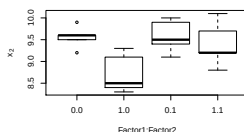
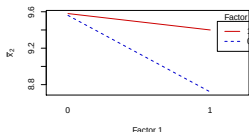
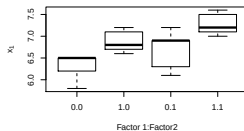
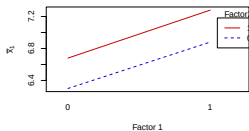
```
> plastic <- read.table('T6-4.dat')
> attach(plastic)
> X <- cbind(V3, V4, V5)

> #Fit two-way MANOVA
> summary(manova(X ~ V1*V2), test = 'Wilks')
```

	Df	Wilks	approx F	num Df	den Df	Pr(>F)
V1	1	0.38186	7.5543	3	14	0.003034
V2	1	0.52303	4.2556	3	14	0.024745
V1:V2	1	0.77711	1.3385	3	14	0.301782
Residuals	16					

Two-way MANOVA example

The interaction effects are not significant. Both factor effects are significant (we could explore this further).



Homework exercise 6.2

Johnson & Wichern exercise 6.31

Peanuts are an important crop in parts of the southern US. In an effort to develop improved plants, crop scientists routinely compare varieties with respect to several variables. The data for one two-factor experiment are given in `T6-17.dat`.

Three varieties were grown at two locations. Three variables were measured:

X_1 = Yield (plot weight)

X_2 = Sound mature kernels (weight in grams – max 250g)

X_3 = Seed size (weight in grams of 100 seeds)

- a) Perform a two-factor MANOVA, testing for variety effect, location effect, and variety-location interaction.
- b) Analyse the residuals from part a). Do the usual MANOVA assumptions appear to be satisfied?